NYANZA ADVENTIST SECONDARY SCHOOL
FORM FOUR PRWMOCK 1 EXAMINATIONS 2025

031/1 **PHYSICS 1**

TIME: 2:30 Hrs

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer **all** questions in sections A and B and two (2) questions from section C.
3. Section A carries sixteen (16) marks, section B carries fifty four (54) marks and section C carries thirty (30) marks.
4. All writings must be in blue or black ink, except drawings which should be in pencil.
5. Cellular phones and any unauthorized materials are not allowed in Examination room.
6. Non-programmable calculators and mathematical tables may be used.
7. Write your Examination Number on every page of your answer booklet(s).
8. Where necessary the following constants may be used:
   (i) Acceleration due to gravity, $g = 10m/s^2$.
   (ii) Density of water, $\rho = 1000\ kgm^{-3}$.
   (iii) Pie, $\pi = 3.14$.
   (iv) Specific latent heat of vaporization of water = $2.3 \times 10^6 Jkg$

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

   (i) The Physics teacher from Changanyikeni Secondary School ordered her students to measure the diameter of a wire to the nearest 0.001cm. As one of the form four students, which of the following apparatus could be precise?
   A. Vernier Caliper
   B. Micrometer screw gauge
   C. Metre rule
   D. Pair of compasses
   E. Engineer Caliper

   (ii) A glass rod is rubbed with silk. As a result, the glass rod becomes positively charged and silk attains an equal negative charge. Explain the application of this practical as a physics student.
   A. Silk gains as many electrons as lost by the glass rod
   B. Silk is a very good conductor and receives all the electrons
   C. Glass rod practically is a good receiver of all the electrons in nature
   D. Glass rod and silk are the best conductors and good receivers
   E. Silk is a bad conductor and can never receive electrons

   (iii) An object was weighed in air by a Professor of Physics and she got 20N, she immersed the object in water and she got 15N and later immersed in liquid L. What was the relative density of that liquid L measured by that Professor?
   A. 5N
   B. 0.4N
   C. 0.4
   D. 2.5
   E. 0.5

   (iv) Basing on the principle of Ohm's law. If the resistance is doubled and the potential difference across it is reduced to one third, what is the ratio of old current to the new current?
   A. $\frac{3}{2}$
   B. $\frac{2}{3}$
   C. $\frac{1}{6}$
   D. $\frac{6}{1}$
   E. $\frac{1}{3}$

(v) It is believed by Physicists that the interior structure of the earth is mainly composed of three major zones. As a form four physics student where does this research base amongst the following?

A. Magma, mantle and the core
B. Lava, crust and magma
C. Hypocenter, crust and the mantle
D. Crust, mantle and the core
E. The core, lava, and hypocenter

(vi) It is practically observed in the laboratory that diffusion occurs more quickly in a gas than compared in a liquid, this perhaps is due to:

A. The liquid contains a layer on its surface
B. The gas contains semi-permeable membrane
C. The gas molecules are small in size compared to the liquid molecules
D. The adhesion is larger than cohesion in gas compared to that in liquid
E. The speed of molecules in gas is greater than in liquid.

(vii) A magnetic needle, free to turn in a vertical plane is suspended first at the earth's south magnetic pole and then at a point on the magnetic equator. The respective angles between the needle and horizontal are:

A. 0° and 0°
B. 70° and 70°
C. 0° and 45°
D. 0° and 90°
E. 90° and 0°

(viii) A typist uses a new carbon paper under her top typing for making a copy of letter. When she held the carbon paper close to a plane mirror, she was able to read the letter, this is because the letter:

A. Forms an image the same size as an object
B. Produces an inverted image
C. Produces a laterally image
D. Forms a virtual image
E. Forms an image behind the mirror.

(ix) The government of Tanzania has stopped all busses carrying very big load on their carrier and people standing in the busses as the vehicle is likely to overturn due to the fact that:

A. It cannot run very fast and reach the destination
B. Its equilibrium is stable
C. Its centre of gravity is low
D. It is more stale and can run fast
E. Its center of gravity is high.

(x) A horizontal electron beam passes between two parallel horizontal plates X and Y with X above Y when a high tension battery is connected with its negative pole to X and positive pole to Y, the beam will be deflected:

A. Upwards
B. Downwards at 45° to the horizontal
C. Upwards at 45° to the horizontal
D. Downwards
E. Sideways.

2. Match the terms of wave in list A with the definitions of wave terms in list B by writing the letter of the corresponding response beside the item number in the answer sheet provided.

	LIST A	LIST B
(i)	Longitudinal waves	A. Sin r over sin i
(ii)	Reverberation	B. Refractive index
(iii)	Snell's law	C. Sound wave
(iv)	Critical angle	D. Angle of incidence for which the angle of refractive is 90°.
(v)	Principle of superposition	E. The resultant displacement at any point is equal to the sum of displacement of different waves.
(vi)	End correction	F. The multiple reflection of sound waves.
		G. Angle of reflection for which the angle of incidence is 90°.
		H. Is the point on top of the tube formed in air

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) While in a practical, a student was required to use a half metre rule and to be balanced at 15cm mark when a load of 0.4N was hanging amongst them. Draw a sketched diagram indicating the arrow of weight of the rule acting through the centre of gravity hence determine the weight of that half metre rule. **[03 marks]**

(b) Students at Moshi secondary school, were using a screw Jack which had a screw pitch of 5mm with an effort arm of 16cm in the laboratory. Their physics teacher asked them first to state two forms of energy in which the energy supplied to screw Jack is finally converted to and lastly to determine the percentage efficiency of that screw Jack, if it needed an effort of 30N to lift a load of 750N. As a form four student give us precise answers. **[06 marks]**

4. (a) Explain three real practical examples in which impulse and momentum play on the important role in day to day life here in Tanzania. **[03 marks]**

(b) A young man of 25 years was driving a 1000kg car at a very high speed and later collided with a truck of 5000kg. During the collision, the truck exerted a force of

10,000 Newton on the car. What can you say on the accelerations of the car and the truck?

[06 marks]

5. (a) Scientists believe with no doubt that it requires less force to keep an object moving at a constant velocity than to start moving. Explain your ideal on this fact? **[03 marks]**

   (b) A Geography teacher was pulling an aluminum block of mass 2.1kg resting on a steel platform. She applied a horizontal force of 15N to the block.

   (i) If the coefficient of limiting friction between the surfaces was 0.61, will the block move?

   (ii) If it moves, what will be its acceleration? **[06 marks]**

6. (a) Triangle law of vector addition is appropriate when adding two vectors, explain this law by using diagrams. **[03 marks]**

   (b) Mrs. Sumaiya was driving car A moving with a velocity of 20m/s while Mrs. Mchanga was driving car B with velocity of 30m/s. Comment on the velocity of car B relative to car A if:

   (i) They are moving in the same direction.

   (ii) They are moving in opposite direction **[03 marks]**

7. (a) It is believed and practically done that Air conditions are strategically placed at a high point in the house and Room heaters on the other hand, are placed on the floor of a Room. As a form four student taking physics explain the hidden agenda for this?

   [04 marks]

   (b) An engineer has one piece of metal of plate to be used in fixing a gap on a metal bridge. If the steel plate has area of $0.8m^2$ at 25^oC, by how much the engineer should raise the temperature of the steel plate for it to fit a gap of $0.804m^2$?
   [Use coefficient of linear expansion of steel as 0.000012^oC^{-1}] **[04 marks]**

8. (a) Force and pressure are common terms used in day to day situations and almost in all cases. Elaborate how they are related and how does force become weight [**03 marks]**

   (b) A rectangular wooden rod of density $200kg/m^3$ has dimensions of 0.3m x 0.5m x 6.0m.

   (i) Calculate the maximum pressure it can exert on the ground and how is it experienced.

   (ii) Calculate the minimum pressure it can exert on the ground, and how can this be observed. **[06 marks]**

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. (a) (i) Faraday's law and Lenz's law are precise laws dealing with induced e.m.f.

   How do they differ when there is magnetic flux linking the coil concerned?

   [03 marks]

(ii) The results of the Faraday's experiments were that the magnitude of the induced e.m.f were related to which factors? Explain in short four factors. **[02 marks]**

(b) An ideal transformer should have an efficiency of 100%. Yet in any practical transformer there are power losses which do not allow 100% efficiency. Explain five of those factors. **[05 marks]**

(c) A step-up transformer has 10,000 turns in the secondary coil and 100 turns in the primary coil. A current of 5.0A flows in the primary circuit when connected to a 12.0Vsupply.

(i) Calculate the voltage across the secondary coil.

(ii) If the transformer has an efficiency of 90%, what is the current in the secondary coil? **[05 marks]**

10. (a) Explain using a diagram how the sensing element of a Geiger counter instrument can detect a single particle of ionizing radiation. **[05 marks]**

(b) Explain shortly four applications of natural radioactive isotopes in this science and technology to Tanzania government. **[05 marks]**

(c) One isotopes of chlorine has the symbol, $^{37}_{17}Cl$

(i) Determine the number of Neutrons in this isotope.

(ii) Calculate the charge, in coulombs, on the nucleus of this isotope. **[05 marks]**

11. (a) Information is usually transmitted in electronic devices in form of signals. Explain with diagrams the two broad classes of information signals. **[05 marks]**

(b) In the modern technological world, Tanzania is not behind, it has seen that electronics is a valuable and ever growing sector. As a physics student explain five plenty of fields where electronics is applied. **[05 marks]**

(c) In a common emitter circuit, calculate the emitter current, I_E of a transistor for which current gain β is 50 and base current I_B is 20μA. **[05 marks]**